Mehrad Ansari

LinkedIn Google Scholar GitHub Website Blog

AI research scientist building autonomous agents and tools for scientific discovery, experience in LLM post-training (SFT/RL), production agent and RAG infrastructure, and AI-driven molecular design across drug and materials discovery.

EDUCATION

Ph.D., Chemical Engineering University of Rochester 2019 - 2023
Thesis: Applications of Physics-informed Machine Learning in Chemical Engineering (Advisor: Andrew D. White)

M.S., Chemical Engineering University of Rochester 2019 - 2021

M.S., Environmental Engineering Missouri University of Science & Technology 2016 – 2018
Thesis: Numerical Modeling of Capillary-driven Flow in Open Microchannels: An Implication of Optimized Wicking Fabric Design

B.S., Chemical Engineering University of Tehran 2010 – 2015
Thesis: Experimental Setup and Optimization for Electro-catalytical Generation of Hydroxyl Radicals in Wastewater Treatment

EXPERIENCE

Senior Scientist
June 2025 – Present
Lila Sciences, Boston, MA

- Designed scientific AI agents with ReAct reasoning, long-term memory, and tool use for long-horizon inverse molecular design tasks, coupling LLM reasoning with physics-informed surrogate models; built stateful code-execution sandboxes for autonomous experimentation.
- Fine-tuned open-weight LLMs (Mistral-7B) with supervised fine-tuning and GRPO (*reinforcement learning from verifiable rewards*) for constrained SMILES molecular generation on EC2 GPU instances.
- Built a scalable scientific-literature RAG service exposed via an MCP endpoint on Kubernetes/AWS; designed evaluation benchmarks (LitQA2) and optimized hybrid BM25+dense retrieval with reranking, benchmarked against PaperQA2.
- Built a therapeutic bioactivity-prediction system for EGFR inhibitor potency (IC50) using ChEMBL data.
- Led and mentored cross-functional teams of PhD-level researchers and engineers across materials and drug discovery.

Staff Scientist
Jan 2024 – June 2025

Acceleration Consortium Research Fellow
June – Dec 2023
Acceleration Consortium, University of Toronto, Toronto, ON

- Lead developer of **dZiner**, an LLM multi-agent framework for interpretable, literature-grounded inverse molecular design. Designed *de novo* small-molecule inhibitors of the WDR5 oncology target; top candidates were synthesized (WuXi AppTec) and experimentally validated via fluorescence-polarization displacement, affinity-selection mass spectrometry, and differential scanning fluorimetry (manuscript under review, *Nature Machine Intelligence*).
- Applied representation learning, Bayesian optimization, and AI agents to accelerate self-driving chemistry-lab campaigns and hit-to-lead optimization with built-in domain knowledge.
- Developed chemistry-informed AI agents (Slack API) and agent-based LLM plugins to accelerate design of materials for CO_2 reduction.
- Built **Eunomia**, an agent-based system for large-scale extraction of materials datasets from scientific literature (*Digital Discovery*).
- Mentored graduate and undergraduate students and co-organized global hackathons on AI for chemistry and materials.

Research Assistant
2019 – 2023
University of Rochester, Rochester, NY

- HuggingFace literature-QA app that answers questions from scientific papers using LLMs — an early retrieval-augmented scientific assistant.
- Developed Peptide.bio, an edge-computing cheminformatics tool for semi-supervised classification of antimicrobial-peptide activity via positive-unlabeled learning with recurrent neural networks.
- Built disease-modeling (py0) and active-learning CFD / symbolic-regression tools (AL-CFD) for scientific simulation.

Energy & Materials ML Research Engineer / Intern
May 2022 – Mar 2023
Toyota Research Institute, Los Altos, CA

- Developed deep-learning architectures to predict degradation of used Li-ion batteries with unknown cycling histories (**US Patent pending**).

Manufacturing Process Modeling Intern
May – Dec 2017
The Goodyear Tire & Rubber Company, Akron, OH

- Modeled and optimized the tire-vulcanization process (phase-change heat transfer) in ANSYS; automated simulation workflows (OPTIMUS, MATLAB) and validated against plant data using adaptive mesh refinement on HPC.

Research Assistant
2016 – 2018

Missouri University of Science & Technology, Rolla, MO

- Numerical modeling of multiphase flow in open microfluidics (ANSYS, STAR-CCM+); developed adaptive mesh-refinement algorithms.

SELECTED PUBLICATIONS & PATENTS (see full list on Google Scholar)

1. **dZiner: Rational Inverse Design of Materials with AI Agents** Project Lead — Spotlight, AI4Mat-NeurIPS (2024), *Nature Machine Intelligence (under review 2026)*
   M. Ansari, *J. Watchorn, C. E. Brown, J. S. Brown*
2. **Agent-based Learning of Materials Datasets from Scientific Literature (Eunomia)** — *Digital Discovery (2024)*
   M. Ansari, *S. M. Moosavi*
3. **Learning Peptide Properties with Positive Examples Only** — *Digital Discovery (2024)*
   M. Ansari, *A. D. White*
4. **32 Examples of LLM Applications in Materials Science and Chemistry** Project co-organizer & team lead — *MLST (2025)*
   Y. Zimmermann, A. Bazgir, A. Al-Feghali, **M. Ansari**, et al.
5. **Kernel-Learning-Assisted Synthesis Condition Exploration for Ternary Spinel** Project co-lead — *Nature Communications Materials (2025)*
   Y. Liu, **M. Ansari**, R. Black, J. Hattrick-Simpers
6. **History-agnostic Battery Degradation Inference** — *Journal of Energy Storage (2024) · US Patent Pending*
   M. Ansari, S. Torrisi, A. Trewartha, S. Sun
7. **Serverless Prediction of Peptide Properties with Recurrent Neural Networks** — *Journal of Chemical Information and Modeling (2023)*
   M. Ansari, A. D. White
8. **Assessment of Chemistry Knowledge in Large Language Models that Generate Code** — *Digital Discovery (2023)*
   A. D. White, G. M. Hocky, H. A. Gandhi, **M. Ansari**, S. Cox, G. P. Wellawatte, S. Sasmal, Z. Yang, K. Liu, Y. Singh, W. J. Peña Ccoa
9. **Bayesian Optimization Hackathon for Chemistry and Materials** Project co-lead — *chemRxiv(2025)*
   S. Baird, **M. Ansari**, Z. Afzal, Q. Ai, et al.
10. **Iterative Symbolic Regression for Learning Transport Equations** — *AIChE Journal (Special Issue on AI 2022)*
    M. Ansari, H. A. Gandhi, D. G. Foster, A. D. White
11. **Inferring Spatial Source of Disease Outbreaks using Maximum Entropy** — *Physical Review E (2022)*
    M. Ansari, D. Soriano-Paños, G. Ghoshal, A. D. White
12. **Simulation-based Inference with Approximately Correct Parameters via Maximum Entropy** — *NeurIPS (2020), MLST (2022)*
    R. Barrett, **M. Ansari**, G. Ghoshal, A. D. White

SELECTED TALKS

- Teaching Computers How to Do Science (Invited) — *IBM Research, Tokyo, Japan* — Feb 2025
- Rational Inverse Design of Materials with Chemist AI Agents — *Spotlight, AI4Mat-NeurIPS 2024, Vancouver, BC* — Dec 2024
- Materials Inverse Design with AI Agents (Invited) — *Trillion Parameter Consortium, Barcelona, Spain* — June 2024
- Multi-modal AI Agents in Materials Discovery — *Accelerate 24, Vancouver, BC* — Aug 2024
- Flexible Automation of Self-driving Labs with Built-in Domain Knowledge (Invited) — *Toyota Research Institute, Los Altos, CA* — Mar 2024
- Positive Unlabeled Learning of Peptide Properties — *Accelerate23, Toronto, ON* — Aug 2023

HONORS & AWARDS

- Acceleration Consortium Research Fellowship, University of Toronto — 2023
- 1st Place — Battery Informatics & ML Kaggle Competition, Materials Research Society — 2022
- 3rd Place (Radical AI) — LLM Hackathon for Materials & Chemistry, Toronto — 2024
- 2nd Place — Bayesian Optimization Hackathon for Chemistry & Materials, Toronto — 2024
- Kwang-Yu and Lee-Chien Wang Fellowship, University of Rochester — 2021
- Earl W. Costich Graduate Fellowship, University of Rochester — 2020
- Recognized Reviewer, Journal of Environmental Chemical Engineering — 2016

TECHNICAL SKILLS

Scientific Software: dZiner · Eunomia · Peptide.bio · MaxEnt · Py0 · AL-CFD · HOOMD-TF

LLM & agents: RAG, multi-agent frameworks, ReAct, MCP, RLHF/GRPO, SFT, eval harnesses (LitQA2)

ML / cheminfo: PyTorch, TensorFlow, JAX, scikit-learn, LiteLLM, vLLM, RDKit; Bayesian optimization, surrogate modeling

Data & DBs: ChEMBL, PubChem, PDB

Infrastructure: AWS, Kubernetes, Ray, HPC/Slurm, Lambda Labs

Languages: Python, JavaScript, HTML/CSS